AGENTIC AI PROOF OF CONCEPT

OMNITRIX

A multi-agent career assistant that finds, validates and tracks opportunities, and tailors resumes without inventing facts

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Domain: Education and career services



THE PROBLEM

OPPORTUNITY SEARCH IS SCATTERED AND RISKY



Students check many job boards. The same role shows up on several platforms, and some listings are expired, closed or have no deadline.



Applications, deadlines and follow-ups live in notes and spreadsheets, so dates get missed.



Tailoring a resume for every role tempts students to overstate skills they have not really used.

WHY THIS DOMAIN

Career search is multi-step, deadline-driven and trust-sensitive. That makes it a good fit for agents that plan, use tools and validate before showing anything.

30

synthetic listings in the demo dataset

9

deliberately tricky records to test validation

0

applications ever submitted automatically

THE SHIFT

FROM MANUAL HUNTING TO COOPERATING AGENTS

TODAY (MANUAL)

- Search each portal separately and copy details into a sheet
- Find duplicates and expired postings only after clicking through
- Track deadlines from memory and forget follow-ups
- Edit the resume by hand for each job, easy to overstate

WITH OMNITRIX

- One request: the Planner picks the steps and filters
- Matching ranks every listing with a score and explanation; the Critic removes duplicates, closed and invalid records
- Follow-up agent suggests actions from recorded dates only
- Resume agents ask before adding any skill; a critic blocks unsupported claims

OPPORTUNITY WORKFLOW

SIX AGENTS, ONE JOB EACH



PLANNER 01

Turns free text into an ordered plan and filters (internship, research, closing soon). Deterministic keyword routing.



DISCOVERY 02

Filters the local dataset by type, mode, location, role, skill and deadline window. Never touches the network; leaves bad rows for the Critic.



MATCHING 03

Scores 0-100: skills 50, role 20, location and mode 15, type 10, deadline 5. Writes a plain-English explanation.



FOLLOW-UP 04

Builds a prioritised action list from recorded deadlines, applied and follow-up dates. Every action is labelled a suggestion.



CRITIC 05

Checks required fields, dates, closed status, duplicates, expiry and source URL, then re-derives each explanation.



TRACKER 06

The only agent that writes to SQLite, and only after an explicit user action. Saving the same job twice is a no-op.

ORCHESTRATION

THE PLANNER DECIDES WHICH STEPS RUN

FIND

“Find internships for me”

Discovery → Matching → Follow-up → Critic

DEADLINES

“Show deadlines this week”

Discovery → Follow-up → Critic

FOLLOW-UPS

“Which applications need follow-up?”

Follow-up → Critic

DECISION POINTS

No candidates found

Return a clear “no match” message with advice

Best score below 25

Flag “no strong match” instead of showing weak results

Critic finds a bad record

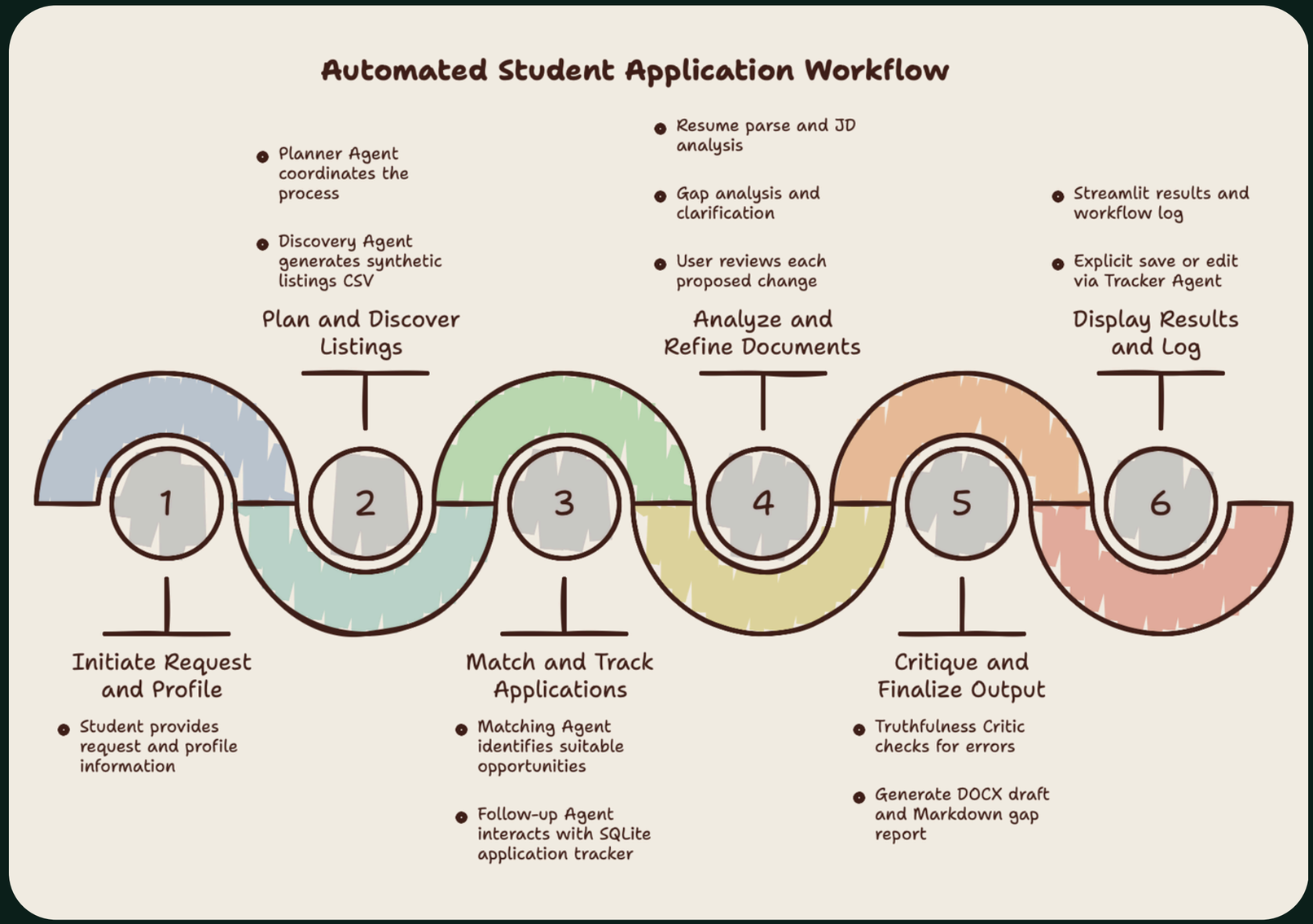
Exclude it and record the reason

Tracker database unavailable

Search results are still shown

ARCHITECTURE

ARCHITECTURE AT A GLANCE



12

agents across two lanes

2

critics: listings and resume truth

1

agent allowed to write data

RESUME TAILORING

ASK FIRST, EDIT SECOND, VERIFY LAST

Parse	PDF, DOCX or text into sections
Analyze JD	Separate required from preferred skills
Gap analysis	Find resume evidence per requirement
Clarify	Ask one question per gap
Tailor	Propose changes that have a source
Truth check	Critic blocks invented claims; errors block export

SAMPLE RUN

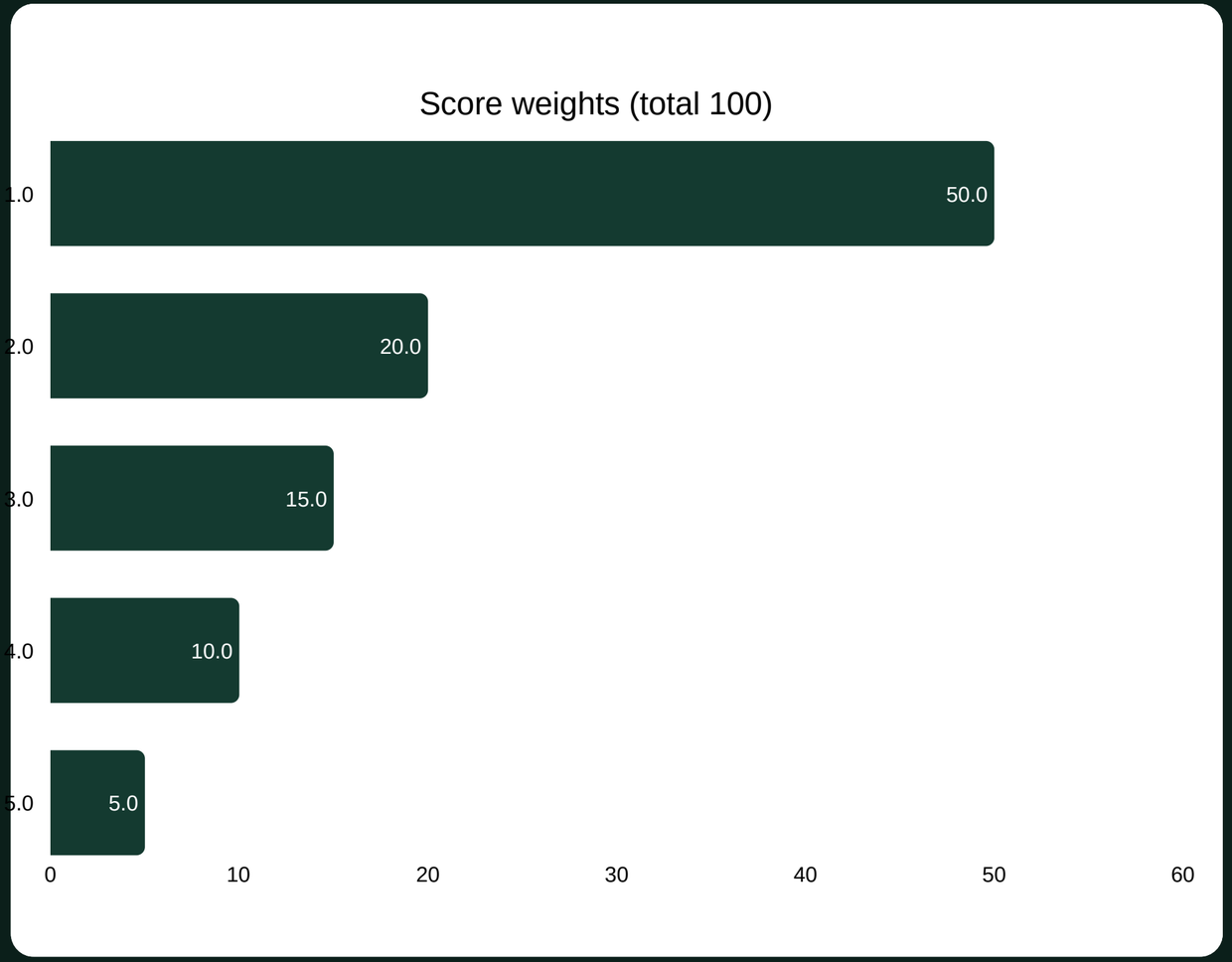
NLP INTERN JOB DESCRIPTION

python	Demonstrated
nlp	Demonstrated
transformers	Demonstrated
sql	Demonstrated
docker	Not found
deployment (preferred)	Mentioned but unclear
git (preferred)	Mentioned but unclear

4 of 7 requirements demonstrated, 3 clarifying questions generated. “Not found” never means “lacks the skill”, so the agent asks.

EXPLAINABLE MATCHING

EVERY SCORE CAN BE TRACED



REAL OUTPUT

SYN-001, AI/ML INTERN

98

“Matching skills: python, machine learning, pytorch, sql. Role: title contains preferred role ‘AI’. Work mode matches (Hybrid); location matches (Pune). Type matches (Internship). Deadline: closes in 5 day(s).”

CRITIC RE-CHECK

Recomputes matched and missing skills from the listing and profile. A record whose explanation disagrees is excluded.

OUTPUT VALIDATION

NINE SEEDED TRAPS, EACH HANDLED BY A RULE

RECORD	PROBLEM	AGENT BEHAVIOUR
SYN-005	Cross-platform duplicate of SYN-001	**Excluded**, grouped under the original
SYN-029	Exact duplicate of SYN-003	**Excluded** as duplicate
SYN-012, SYN-024	No deadline	**Kept**, flagged “Deadline unknown”
SYN-008, SYN-019	Deadline already passed	**Kept**, flagged expired, ranked below the rest
SYN-015	No source URL	**Kept**, flagged “Source unavailable”
SYN-021	Listing status Closed	**Excluded**
SYN-027	Invalid date format	**Excluded**

Nothing is silently dropped. Every excluded record keeps a reason, shown in the UI.

RESULTS

RESULTS FROM A REAL RUN OF THE APP

30 / 30

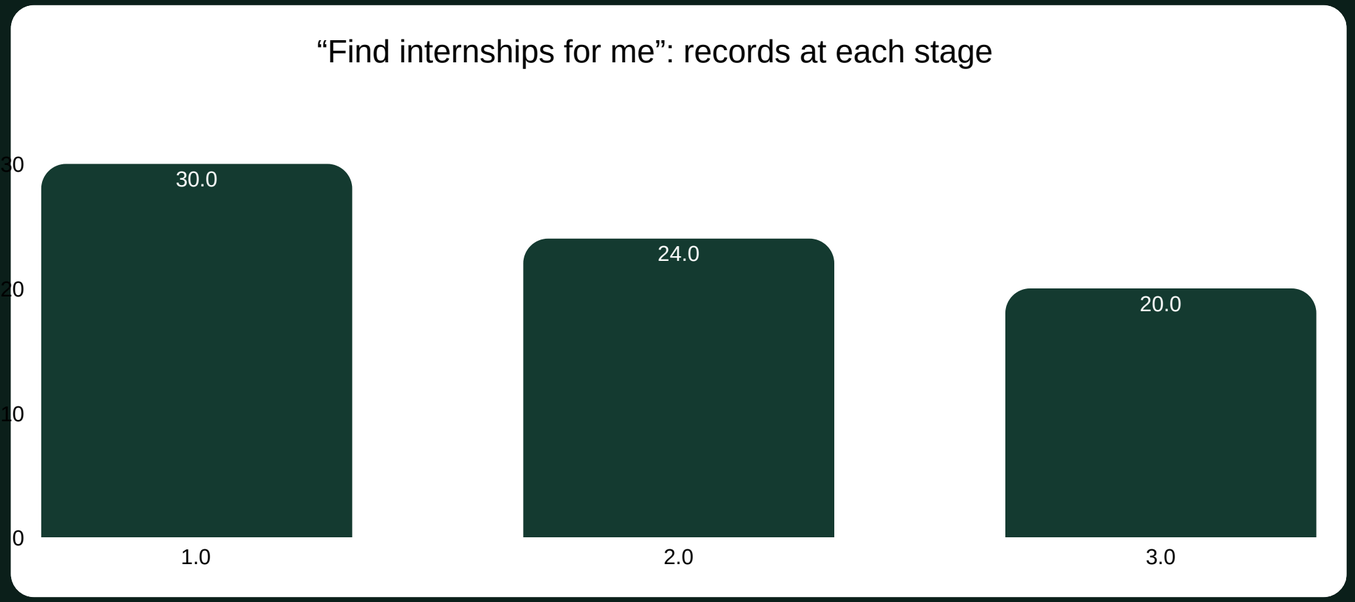
automated tests pass

4

bad records caught out of 24 internship matches

4 OF 7

JD requirements already shown in the sample resume



The 4 excluded

2 duplicates, 1 closed listing, 1 invalid date

4 suggested actions

Produced by the same run

Fixed demo date

2026-10-15, so results repeat exactly

Synthetic data. Results show the workflow is correct, not real-world job-matching accuracy.

WHY THIS IS AGENTIC, NOT A CHATBOT

PLAN. ACT. CHECK. EXPLAIN.

Five properties that separate a team of agents from a chat window.



TASK PLANNING

Planner converts a request into an ordered plan with inferred filters.



EXECUTION WITH TOOLS

Agents run filtering, scoring, SQLite reads and writes, and DOCX export.



DECISION-MAKING

Routing, no-match thresholds, and accept, edit or reject choices on every resume change.



VALIDATION

Two critics: one for listings and explanations, one for resume truthfulness.



COLLABORATION

Agents share state through the orchestrator, and every step lands in a log trail.

WHAT'S NEXT

LIMITATIONS AND FUTURE SCOPE

LIMITATIONS

- Synthetic data only: 30 listings, no live job feeds
- Keyword-based matching and rule-based agents, no LLM reasoning
- No employer status checks; follow-ups are suggestions
- Resume export is DOCX only (no PDF)

FUTURE SCOPE

- Live job sources through official APIs, with user consent
- Semantic matching with embeddings
- Local LLM Planner and bullet rewriter through the existing hooks, still checked by the critic
- Calendar and email reminders, PDF export

LIVE DEMO

LIVE DEMO WALKTHROUGH

```
$ pip install -r requirements.txt
$ streamlit run app.py
```

1

SIDEBAR

Set the student profile, click “Seed demo data”

2

FIND OPPORTUNITIES

Run “Find internships for me” and open a score explanation

3

AGENT WORKFLOW TAB

Show the log trail and excluded records with reasons

4

DEADLINES & FOLLOW-UPS

Show the suggested actions

5

RESUME TAILOR

Paste the sample JD, answer the questions, review changes, try editing in an unsupported skill and watch export get blocked